

1	2	3	4	5	6	Sum
/20	/20	/20	/20	/20	/20	/100

Name _____

June 6, 2008 Friday

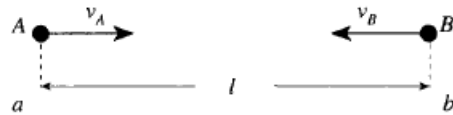
Department _____

SID _____

University Physics
Practice Final Examination
Kuang Yaming Honors School, Nanjing University

Select five out of following six problems.

1. (20 pts.) A person on Earth observes two rocket ships moving directly toward each other and colliding. At time $t = 0$ in the Earth frame, the Earth observer determines that rocket A, traveling to the right at $v_A = 0.8c$, is at point a , and rocket B is at point b , traveling to the left at $v_B = 0.6c$. They are separated by a distance $l = 4.2 \times 10^8 \text{ m}$.



(a) In the Earth frame, how much time will pass before the rockets collide?

(b) How fast is rocket B approaching in A's frame? How fast is rocket A approaching in B's frame?

(c) How much time will elapse in A's frame from the time rocket A passes point a until collision? How much time will elapse in B's frame from the time rocket B passes point b until collision?

2. (20 pts.) A gas obeys the equation of state

$$p = \frac{T}{V} + \frac{B(T)}{V^2}$$

where $B(T)$ is a function of temperature only. The gas is expanded isothermally at temperature T from the volume V_0 to the volume $V_1 = 2V_0$. The process is reversible.

(a) Find the work done in the expansion.

(b) Find the heat absorbed in the expansion.

3. (20 pts.) The equation of state for one mole of Van der Waals gas can be written as

$$\left(p + \frac{a}{V^2} \right) (V - b) = RT$$

and the internal energy for the Van der Waals gas is given by

$$U = U_0 + C_V T - \frac{a}{V},$$

where C_V is assumed to be constant.

(a) Determine the adiabatic equation of the one mole of Van der Waals gas.

(b) If the gas expands freely into the vacuum from the volume V_1 to the volume V_2 , what is the temperature change in this process?

4. (20 pts.) If 20 grams of ice at temperature of -10°C is changed into water of 10°C at 1 atm., what is the change of the entropy? (The specific heat of the ice is $c_{pi} = 2.1 \text{ J/g} \cdot \text{K}$ and the specific heat for the water is $c_{pw} = 4.2 \text{ J/g} \cdot \text{K}$. The latent heat absorbed during the ice-water phase transition is $L = 333 \text{ J/g}$.)

5. (20 pts.) There is a substance of n moles. When the substance expands from a volume V_0 to a volume V at constant temperature T_0 , the work done by it is

$$W = nRT_0 \ln \frac{V}{V_0}.$$

And the entropy of the substance is given by

$$S = nR \frac{V_0}{V} \left(\frac{T}{T_0} \right)^x$$

where x is a fixed constant.

(a) Calculate the Helmholtz free energy of the substance.

(b) Find the equation of state of the substance.

6. (20 pts.) A strip of rubber has length l when subject to a tensile force f . If the volume change due to the extension of the strip can be neglected, the differential of the internal energy is then

$$dU = TdS + fdl .$$

(a) Show that

$$\left(\frac{\partial U}{\partial l} \right)_T = f - T \left(\frac{\partial f}{\partial T} \right)_l .$$

(b) Show that the small temperature rise ΔT that takes place in a slow adiabatic stretching of the strip from the natural length l_0 is given by

$$\frac{\Delta T}{T} = \int_{l_0}^l \frac{1}{C_l} \left(\frac{\partial f}{\partial T} \right)_l dl$$

where C_l is the heat capacity of the strip at constant length l .